

CSE 280 Challenge Set 01 - Solutions

(c) BYU-Idaho

Question 1

Let the following statements be given:

- a = "You used the pool in the afternoon."
- b = "You cleaned up after lunch."
- c = "You must clean up after dinner."

Part 1

Use connectives (operators) to translate the following statement into a compound proposition:

"If you used the pool in the afternoon and you didn't clean up after lunch, then you must clean up after dinner."

Answer: $(a \wedge \neg b) \rightarrow c$

Part 2

Construct a truth table for the compound proposition you found in Part 1. The eight rows of your table should correspond to the eight different possibilities for a , b , and c .

Answer:

a	b	c	$a \wedge \neg b$	ANS
T	T	T	F	T
T	T	F	F	T
T	F	T	T	T
T	F	F	T	F
F	T	T	F	T
F	T	F	F	T
F	F	T	F	T
F	F	F	F	T

Part 3

Suppose that the statement given in part 1 is false. What must be true about your pool usage and cleanup duties? Explain how to justify your answer using the truth table.

Answer: The pool was used and neither the lunch nor the dinner was cleaned up.

Question 2

Use a truth table to prove that the statement $[(p \vee r) \wedge (\neg p)] \rightarrow r$ is always true or always false, no matter what p and r are. What do we call this type of statement?

Answer: A tautology

p	r	$p \vee r$	$\neg p$	$(p \vee r) \wedge (\neg p)$	ANS
T	T	T	F	F	T
T	F	T	F	F	T
F	T	T	T	T	T
F	F	F	T	F	T

Question 3

Let s be the following statement: "If it's raining, then the ground is wet."

Part 1

Give the inverse of s . Is this the same as s ?

Answer: If its not raining, then the ground is not wet. -- No

Part 2

Give the converse of s . Is this the same as s ?

Answer: If the ground is wet, then it is raining. -- No

Part 3

Give the contrapositive of s . Is this the same as s ?

Answer: If the ground is not wet, then its not raining. -- Yes

Question 4

Determine if the following propositions written in English are True or False:

- If 2 is even, then 5 is prime. - True ($T \rightarrow T = T$)
- If 3 is even, then 6 is prime. - True ($F \rightarrow F = T$)
- If 5 is odd, then 8 is prime. - False ($T \rightarrow F = F$)

- If 8 is odd, then 11 is prime. - True ($F \rightarrow T = T$)
- 10 is even if and only if 4 is prime - False ($T \leftrightarrow F = F$)
- 11 is even if and only if 6 is prime - True ($F \leftrightarrow F = T$)

Question 5

Simplify the following using a truth table:

$$\neg(A \rightarrow B) \vee \neg(A \vee B)$$

Answer: $\neg B$

Question 6

The following "ugly-looking" multi-line compound proposition is a Tautology. Can you explain why without using a truth table. Hint: Refer back to Question 3 above.

$$((A \rightarrow B) \leftrightarrow (\neg B \rightarrow \neg A)) \wedge ((B \rightarrow A) \leftrightarrow (\neg A \rightarrow \neg B)) \wedge$$

$$((B \rightarrow C) \leftrightarrow (\neg C \rightarrow \neg B)) \wedge ((C \rightarrow B) \leftrightarrow (\neg B \rightarrow \neg C)) \wedge$$

$$((A \rightarrow C) \leftrightarrow (\neg C \rightarrow \neg A)) \wedge ((C \rightarrow A) \leftrightarrow (\neg A \rightarrow \neg C))$$

Answer: Look at one of the 6 things that are $\wedge$ 'ed together. For example:

$$((A \rightarrow B) \leftrightarrow (\neg B \rightarrow \neg A))$$

Note that the right hand side of the biconditional is the contrapositive of the left hand side. Since the contrapositive is equivalent to the original conditional, then we can say this one part is equal to *True*. The same thing applies to the other 5. We are $\wedge$ 'ing six *True* values. Therefore, it's always True (a tautology).